

tf.compat.v1.sparse_segment_sum Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/sparse_segment_sum

Computes the sum along sparse segments of a tensor.

Function Signatures

```
tf.compat.v1.sparse_segment_sum( data, indices, segment_ids,  
name=None, num_segments=None, sparse_gradient=False )
```

Code Examples

```
tf.compat.v1.sparse_segment_sum(  
data,  
indices,  
segment_ids,  
name=None,  
num_segments=None,  
sparse_gradient=False  
)
```

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data,  
indices,  
segment_ids,  
name=None,  
num_segments=None,  
sparse_gradient=False  
)
```

```
c = tf.constant([[1,2,3,4], [-1,-2,-3,-4], [5,6,7,8]])  
# Select two rows, one segment.  
tf.sparse.segment_sum(c, tf.constant([0, 1]), tf.constant([0, 0]))  
# => [[0 0 0 0]]  
# Select two rows, two segment.  
tf.sparse.segment_sum(c, tf.constant([0, 1]), tf.constant([0, 1]))  
# => [[ 1  2  3  4]  
#      [-1 -2 -3 -4]]  
# With missing segment ids.  
tf.sparse.segment_sum(c, tf.constant([0, 1]), tf.constant([0, 2]),  
num_segments=4)  
# => [[ 1  2  3  4]  
#      [ 0  0  0  0]  
#      [-1 -2 -3 -4]  
#      [ 0  0  0  0]]  
# Select all rows, two segments.  
tf.sparse.segment_sum(c, tf.constant([0, 1, 2]), tf.constant([0, 0,  
1]))  
# => [[0 0 0 0]  
#      [5 6 7 8]]  
# Which is equivalent to:  
tf.math.segment_sum(c, tf.constant([0, 0, 1]))
```

```

c = tf.constant([[1,2,3,4], [-1,-2,-3,-4], [5,6,7,8]])
# Select two rows, one segment.
tf.sparse.segment_sum(c, tf.constant([0, 1]), tf.constant([0, 0]))
# => [[0 0 0 0]]
# Select two rows, two segment.
tf.sparse.segment_sum(c, tf.constant([0, 1]), tf.constant([0, 1]))
# => [[ 1  2  3  4]
#      [-1 -2 -3 -4]]
# With missing segment ids.
tf.sparse.segment_sum(c, tf.constant([0, 1]), tf.constant([0, 2]),
num_segments=4)
# => [[ 1  2  3  4]
#      [ 0  0  0  0]
#      [-1 -2 -3 -4]
#      [ 0  0  0  0]]
# Select all rows, two segments.
tf.sparse.segment_sum(c, tf.constant([0, 1, 2]), tf.constant([0, 0,
1]))
# => [[0 0 0 0]
#      [5 6 7 8]]
# Which is equivalent to:
tf.math.segment_sum(c, tf.constant([0, 0, 1]))

```

Documentation

Computes the sum along sparse segments of a tensor.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.sparse.segment_sum`

Read the section on
segmentation
for an explanation of segments.

Like `tf.math.segment_sum`, but `segment_ids` can have rank less than data's first dimension, selecting a subset of dimension 0, specified by indices. `segment_ids` is allowed to have missing ids, in which case the output will be zeros at those indices. In those cases `num_segments` is used to determine the size of the output.

For example:

`## Returns`